Housing Conditions and Reliance on Polluting Cooking Fuels among Electrified Households in Ghana: Linking Sustainable Development Goals 7 and 11

Abstract

Across sub-Saharan Africa, progress in electricity access has outpaced clean cooking, although the Sustainable Development Goals (SDGs) monitor the two separately. This paper examines whether housing conditions are associated with this electrification–cooking gap in Ghana. Linking the 2021 Population and Housing Census to district indicators, multilevel logistic models are estimated for 661,784 electrified households across 261 districts. Among these, 53.5 per cent use polluting primary cooking fuels. After adjustment for socioeconomic characteristics and district electricity coverage, household non-energy housing deprivation is associated with higher odds of polluting-fuel use (odds ratio [OR] = 2.08), while greater district housing adequacy is associated with lower odds (OR = 0.55). Both associations are evident in urban and rural models, while cooking-space associations differ by settlement. The findings indicate that clean cooking is linked to housing and settlement conditions, supporting integrated assessment of SDG 7 and housing conditions relevant to SDG 11.

Keywords: clean cooking; electricity access; housing conditions; energy poverty; SDG interactions; Ghana